

Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models ¹
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1 Objective

To implement linear and regularized regression models for predicting a continuous target variable (the sanctioned loan amount), evaluate their performance using multiple metrics, visualize model behaviour, and analyze overfitting, underfitting and bias–variance characteristics.

2 Problem Statement

Given the Kaggle / Analytics Vidhya loan prediction dataset (592 applicants after dropping rows with a missing target, 10 raw features covering applicant demographics, income and credit history), the task is to predict the continuous **LoanAmount** sanctioned to each applicant, a supervised **regression** problem. Four linear model families (Linear, Ridge, Lasso and Elastic Net Regression) are trained and compared. Beyond finding the best-performing model, the goal is to understand how regularization strength and penalty type (L1 vs. L2) shape the bias–variance trade-off on this small, skewed dataset.

3 Dataset Description

Dataset Name	Loan Prediction Dataset
Dataset Source	Kaggle / Analytics Vidhya Loan Prediction Practice Problem
Number of Samples (after dropping rows with missing target)	592
Target Variable	LoanAmount (continuous, in thousands)
Raw Feature Count	10 (5 numeric, 5 categorical)
Encoded Feature Count	11 (after one-hot encoding of categorical variables)
Missing Values (raw)	Gender: 13, Married: 3, Dependents: 15, Self_Employed: 32, LoanAmount: 22, Loan_Amount_Term: 14, Credit_History: 50
Train-Test Split	80% train (473 samples) / 20% test (119 samples)

¹GitHub Repository: <https://github.com/blackscythe123/Machine-Learning-course>

4 Exploratory Data Analysis

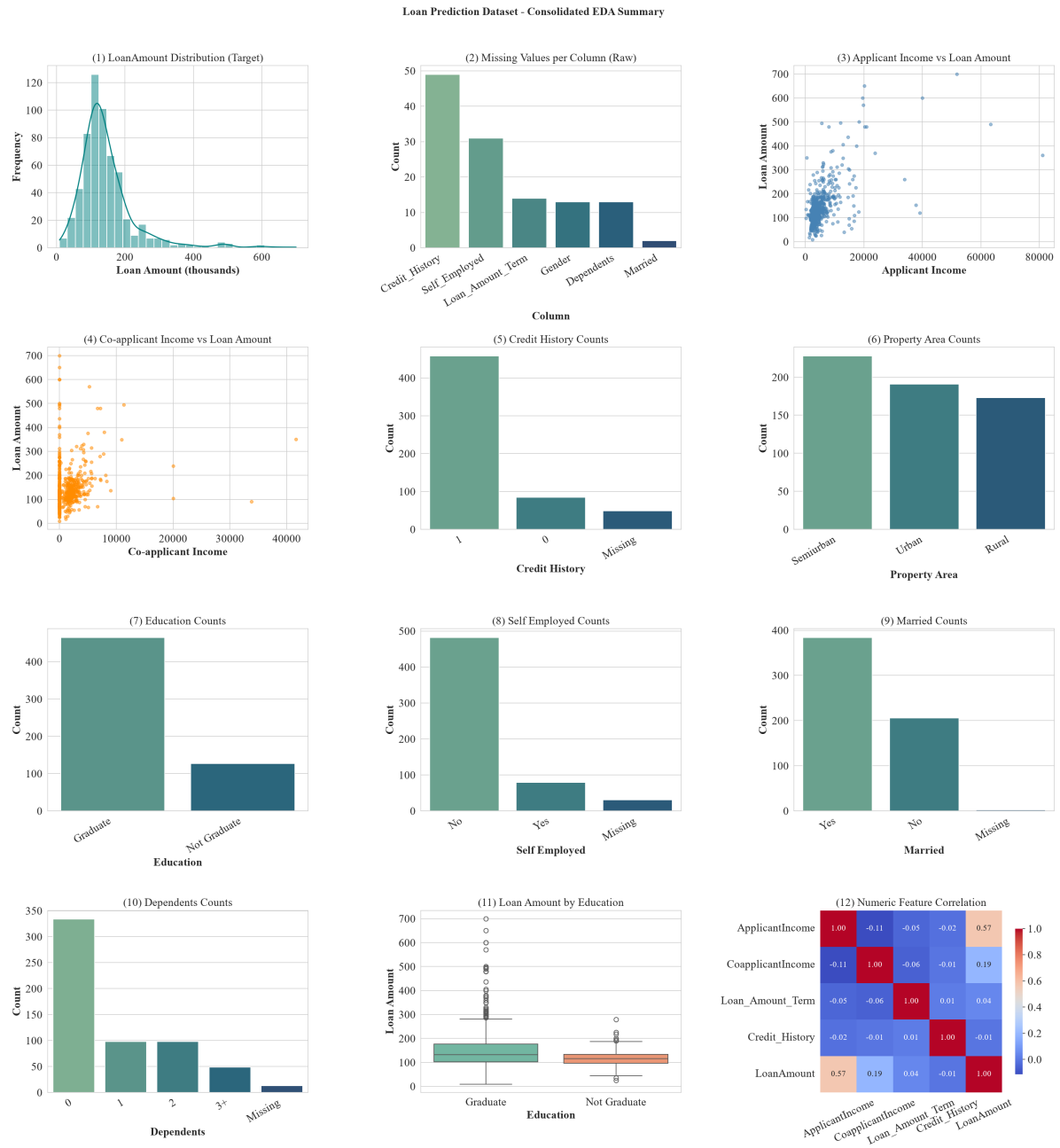


Figure 1: Consolidated exploratory data analysis (12 subplots, single page), on the raw, target-null-dropped, pre-encoding data: (1) `LoanAmount` target distribution; (2) missing-value counts by column; (3–4) `ApplicantIncome` and `CoapplicantIncome` vs. loan amount scatters; (5–10) count plots of `Credit_History`, `Property_Area`, `Education`, `Self_Employed`, `Married` and `Dependents`; (11) `LoanAmount` boxplot by `Education`; (12) correlation heatmap of the numeric features. Exported at 600 DPI as `figures/00_eda.summary.eps` and reproduced here as PNG.

Inference (subplot 1, target distribution): `LoanAmount` is strongly right-skewed, with most applicants sanctioned amounts in the 100–150 (thousand) range and a long

tail of a few large loans. This skew is one reason plain linear models struggle to fit the tail well and partly explains the relatively modest R^2 obtained below.

Inference (subplots 3–4, income scatters): Applicant income shows a weak positive trend with loan amount, with substantial scatter, particularly a handful of high-income outliers that pull the linear fit without necessarily corresponding to proportionally larger loans. This weak, noisy relationship anticipates the limited predictive power ($R^2 \approx 0.1$ – 0.4) observed for every model.

Inference (subplot 2, missing values): `Credit_History` has by far the most missing values (50), followed by `Self_Employed` (32), `Loan_Amount_Term` (14), `Gender` (13), `Dependents` (13) and `Married` (2), motivating the mode/median imputation strategy in the next section.

Inference (subplots 5–10, categorical counts, and subplot 12, correlation): The categorical features are each dominated by one level (e.g. most applicants have `Credit_History`=1, are male and married), which is why one-hot encoding with `drop_first=True` keeps the encoded feature count manageable (11 features from 10 raw). The correlation heatmap confirms `ApplicantIncome` and `CoapplicantIncome` are the only numeric features with a non-trivial linear relationship to `LoanAmount`.

5 Data Preprocessing

- **Target rows with missing `LoanAmount`** (22 rows) were dropped, since imputing the prediction target itself would bias evaluation.
- **Categorical missing values** (`Gender`, `Married`, `Dependents`, `Self_Employed`) were imputed with the column mode.
- **Numeric missing values** (`Loan_Amount_Term`, `Credit_History`) were imputed with the column median, which is robust to the outliers visible in these columns.
- `Dependents` category “3+” was mapped to the integer 3 so the column could be treated as ordinal/numeric.
- **Categorical encoding:** `Gender`, `Married`, `Education`, `Self_Employed` and `Property_Area` were one-hot encoded (`drop_first=True` to avoid the dummy-variable trap).
- **Feature scaling:** All 11 encoded features were standardized (`StandardScaler`, fit on the training split only) so that the regularization penalty in Ridge/Lasso/Elastic Net is applied fairly across features of different natural scale (income figures span thousands; `Credit_History` is 0/1).

6 Machine Learning Algorithm

Four linear model families were trained and compared:

- **Linear Regression** – ordinary least squares, no penalty, serves as the baseline.
- **Ridge Regression** – L2 penalty $\lambda \sum w_i^2$, shrinks coefficients smoothly.
- **Lasso Regression** – L1 penalty $\lambda \sum |w_i|$, can shrink coefficients exactly to zero (feature selection).

- **Elastic Net** – combines L1 and L2 penalties via the `l1_ratio` parameter.

All four minimize a squared-error loss; the three regularized variants add a penalty term controlled by α (and, for Elastic Net, the L1/L2 mixing ratio `l1_ratio`) that shrinks the coefficient vector w toward zero. **Advantages:** Fast to train (closed-form or convex optimization), directly interpretable coefficients, and, for Ridge/Lasso/Elastic Net, a single regularization strength parameter gives explicit control over overfitting. **Limitations:** All four assume a linear relationship between features and target and cannot capture non-linear or interaction effects unless such terms are engineered explicitly; Lasso’s hard feature elimination can also discard features that carry a small but real signal.

7 Experimental Setup

Python Version	3.12.3
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Windows 11
Hardware	Intel Core Ultra 9, NVIDIA RTX 5060

8 Hyperparameter Selection

Ridge, Lasso and Elastic Net were each tuned with `GridSearchCV` using 5-fold KFold cross-validation, optimizing R^2 , over the search space specified in the lab manual.

Table 1: Hyperparameter Tuning Summary

Model	Best Parameters	Best CV R^2
Ridge Regression	$\alpha = 100$	0.3714
Lasso Regression	$\alpha = 10$	0.3311
Elastic Net Regression	$\alpha = 1, l1_ratio = 0.8$	0.3710

9 Results

Table 2: Cross-Validation Performance ($K = 5$, Training Set)

Model	MAE	MSE	RMSE	R^2
Linear Regression	43.06	5506.90	74.21	0.3110
Ridge Regression	44.07	4932.95	70.23	0.3714
Lasso Regression	46.23	5281.51	72.67	0.3311
Elastic Net Regression	43.85	4939.55	70.28	0.3710

Table 3: Test Set Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	36.52	3352.43	57.90	0.0906
Ridge Regression	37.37	3175.95	56.36	0.1385
Lasso Regression	37.80	3176.80	56.36	0.1382
Elastic Net Regression	37.32	3163.59	56.25	0.1418

Table 4: Coefficient Comparison (Top 8 Features)

Feature	Linear	Ridge	Lasso	Elastic Net
ApplicantIncome	54.03	43.88	44.98	43.76
CoapplicantIncome	22.30	17.22	11.79	16.76
Married_Yes	9.43	7.87	1.74	7.51
Loan_Amount_Term	7.71	5.64	0.00	4.98
Self_Employed_Yes	6.46	6.18	0.00	5.66
Education_Not Graduate	-6.38	-6.81	0.00	-6.23
Property_Area_Urban	-5.47	-4.12	0.00	-3.01
Dependents	4.98	5.72	0.00	5.28

10 Result Visualization

Predicted vs. Actual Values

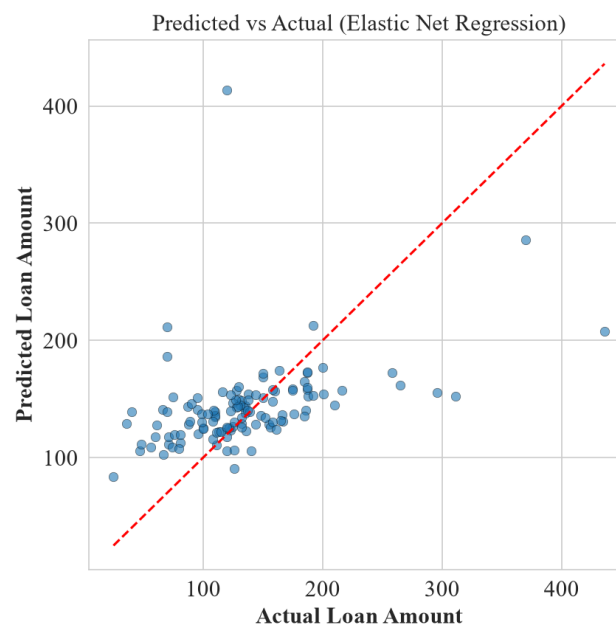


Figure 2: Predicted vs. actual loan amount for the best test-set model (Elastic Net).

Inference: Points cluster loosely around the diagonal for typical loan amounts but the model systematically under-predicts the largest loans, visible as points falling well below

the red reference line at the top-right, a direct consequence of the right-skewed target distribution and the linear models' inability to represent the non-linear tail behaviour.

Residual Plot

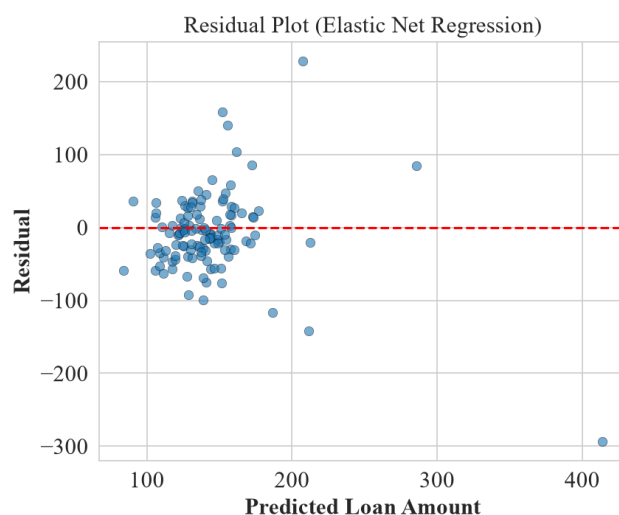


Figure 3: Residuals vs. predicted values (Elastic Net).

Inference: Residuals fan out and become more positive (under-prediction) as the predicted loan amount increases, a classic sign of heteroscedasticity linked to the skewed target; a log-transform of `LoanAmount` would likely stabilize this variance and is a natural next step beyond the scope of this experiment.

Training vs. Validation Error

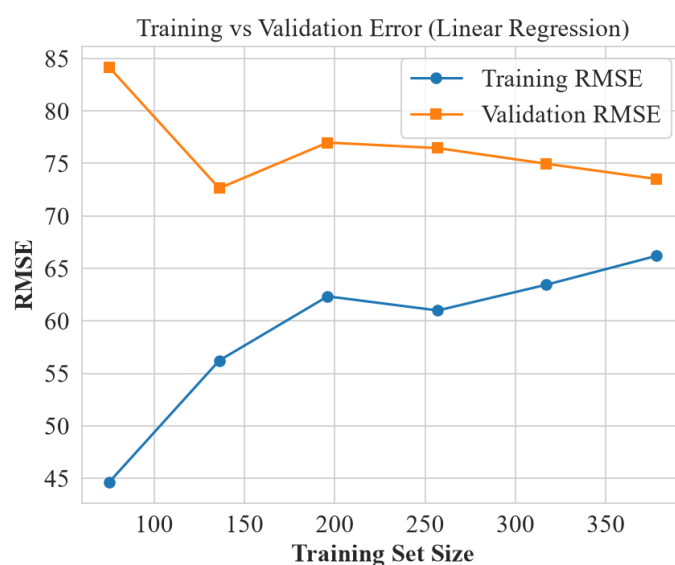


Figure 4: Training vs. validation RMSE as training set size increases (Linear Regression).

Inference: Both curves converge toward a similar, fairly high RMSE as the training set grows, with only a small persistent gap. This pattern is typical of **high bias, low variance**: the linear model is not overfitting (the gap is small), but it is also not able to drive validation error much lower, meaning more data alone would not help much; more informative features or a non-linear model would.

Coefficient Comparison

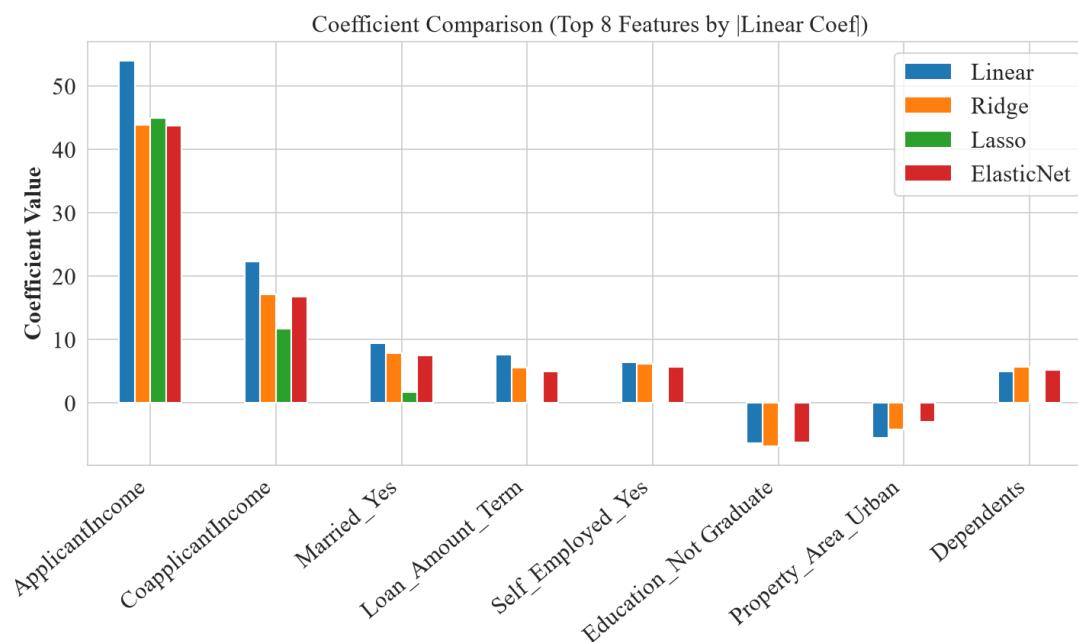


Figure 5: Coefficient magnitudes for the top 8 features, across all four models.

Inference: ApplicantIncome and CoapplicantIncome carry by far the largest coefficients in every model, confirming they are the dominant predictors. Lasso drives Loan_Amount_Term, Self_Employed_Yes, Education_Not Graduate, Property_Area_Urban and Dependents coefficients to exactly zero at its tuned $\alpha = 10$, performing genuine feature selection, while Ridge and Elastic Net shrink these same coefficients toward zero without eliminating them.

Test Set Performance Comparison



Figure 6: R^2 and RMSE for all four models on the held-out test set.

11 Discussion

Overfitting and Underfitting Analysis

Every model's cross-validation R^2 (0.31–0.37, computed on the training folds) is noticeably *higher* than its held-out test R^2 (0.09–0.14). At first glance this looks like overfitting, but the gap here is better explained by the small dataset size (592 samples total, only 119 in the held-out test set): with so few test points, the R^2 estimate itself has high variance, and a handful of large, hard-to-predict loan amounts landing in the test split can noticeably drag test R^2 down independent of model quality. The learning curve (Figure 4) supports this reading: training and validation RMSE are close together and both plateau, which is the signature of **high bias / underfitting**, not high variance / overfitting. Regularization strength reflects this: the best-performing models (Ridge and Elastic Net) both selected comparatively *large* α , i.e. strong regularization, but the improvement over unregularized Linear Regression is modest (CV R^2 0.311 $\rightarrow$ 0.371), consistent with a model that is bias-limited rather than variance-limited. After tuning, generalization (as measured by the CV/test R^2 gap for Ridge and Elastic Net) is not meaningfully better than the untuned baseline, reinforcing that the ceiling here is the information content of the available features, not overfitting.

Bias-Variance Analysis

- **Linear Regression** sits at the high-bias end: it has zero built-in capacity to ignore noisy or weak features, so every feature (including `Loan_Amount_Term` and `Dependents`, whose true effect is likely small) gets a non-zero coefficient, but its overall explanatory power is still limited by the weak linear relationship between features and target.

- **Ridge and Elastic Net** reduce variance by shrinking all coefficients toward zero, which is why they achieve the best cross-validated R^2 ; they trade a small amount of bias for a larger reduction in variance, the classic favourable bias-variance trade at moderate regularization strength.
- **Lasso's feature sparsity effect** is the most visible in Table 4: at its tuned $\alpha = 10$, five of the eight shown coefficients are driven to exactly zero, leaving only `ApplicantIncome`, `CoapplicantIncome` and a small `Married.Yes` effect. This aggressive sparsity slightly *increases* bias relative to Ridge (Lasso's CV R^2 of 0.331 is the lowest of the three regularized models), suggesting that some of the zeroed-out features (e.g. `Credit.History` related dummies, `Education`) do carry a small amount of real signal that Ridge and Elastic Net retain.

12 Conclusion

Elastic Net Regression gave the best held-out test performance (RMSE 56.25, R^2 0.1418) and was statistically tied with Ridge for the best cross-validated R^2 (0.3710 vs. 0.3714), while offering Lasso-like sparsity on some features through its $l1_ratio = 0.8$ setting. Given the small dataset and the loan amount's skewed distribution, none of the four linear models achieves strong absolute predictive power: this reflects a genuine ceiling in the available features (largely income and demographic fields) rather than a modelling failure, and is itself a useful finding: practitioners working with this dataset should expect that accurately predicting the exact sanctioned amount requires additional features (e.g. property value, existing liabilities) beyond what is provided here. Among the four models, **Elastic Net** is recommended as the best trade-off between accuracy, coefficient interpretability and robustness to the correlated income features, since it inherits Ridge's stability while still zeroing out the least useful predictors.

13 References

1. Scikit-learn Developers, "Linear Models," Scikit-learn Documentation. [Online]. Available: https://scikit-learn.org/stable/modules/linear_model.html
2. Scikit-learn Developers, "Tuning the hyper-parameters of an estimator," Scikit-learn Documentation. [Online]. Available: https://scikit-learn.org/stable/modules/grid_search.html
3. Loan Prediction Dataset, Analytics Vidhya / Kaggle Loan Prediction Practice Problem.